

Лабораторная работа №2 (Docker)

Выполнил: студент группы 453501. Кончевич М.В.

Лабораторная работа выполнялась на ОС: MacOS

Задание:

1. Установить Docker (Docker Desktop)
2. Выполнить базовую настройку:

```
[miroslavmeow@MacBook-Air-Miroslav ~ % docker -v
Docker version 28.5.1, build e180ab8
[miroslavmeow@MacBook-Air-Miroslav ~ % docker-compose -v
Docker Compose version v2.40.0-desktop.1
miroslavmeow@MacBook-Air-Miroslav ~ %
```

2. Изучите простейшие консольные команды и возможности Docker Desktop (см. лекцию), создать собственный контейнер docker/getting-started, открыть в браузере и изучить tutorial.

```
Common Commands:
run          Create and run a new container from an image
exec         Execute a command in a running container
ps           List containers
build        Build an image from a Dockerfile
pull         Download an image from a registry
push         Upload an image to a registry
images       List images
login         Log in to a registry
logout        Log out from a registry
search        Search Docker Hub for images
version       Show the Docker version information
info         Display system-wide information
```

```
miroslavmeow@MacBook-Air-Miroslav ~ % docker run -d -p 80:80 docker/getting-started
Unable to find image 'docker/getting-started:latest' locally
latest: Pulling from docker/getting-started
14f901bbf056: Pull complete
261da4162673: Pull complete
a60aada4c44a: Pull complete
33a28b928e89: Pull complete
d0193f05f10f: Pull complete
a879581b8e12: Pull complete
476bb2a1cc22: Pull complete
fa3f58a317be: Pull complete
2f61404bb4b8: Pull complete
Digest: sha256:d79336f4812b6547a53e735480dde67f8f8f7071b414fbd9297609ffb989abc1
Status: Downloaded newer image for docker/getting-started:latest
da90c8f59dc933c2ad66900e372e3eeba8a63d45932c2419f2c6ac1bb4cf7064
```

```
miroslavmeow@MacBook-Air-Miroslav ~ % docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
da90c8f59dc9   docker/getting-started              "/docker-entrypoint..." 57 seconds ago Up 57 seconds 0.0.0.0:80->80/tcp,
[::]:80->80/tcp magical_cannon
4a69fc981aef   welcome-to-docker                  "docker-entrypoint.s..." 6 minutes ago  Up 6 minutes  3000/tcp
vigilant_diffie
```

docker Labs

Getting Started

Search

docker/getting-started

3.0k Stars · 6.4k Forks

Getting Started

Getting Started

Our Application

Updating our App

Sharing our App

Persisting our DB

Using Bind Mounts

Multi-Container Apps

Using Docker Compose

Image Building Best Practices

What Next?

Getting Started

The command you just ran

Congratulations! You have started the container for this tutorial! Let's first explain the command that you just ran. In case you forgot, here's the command:

```
docker run -d -p 80:80 docker/getting-started
```

You'll notice a few flags being used. Here's some more info on them:

- d - run the container in detached mode (in the background)
- p 80:80 - map port 80 of the host to port 80 in the container
- docker/getting-started - the image to use

Pro tip

You can combine single character flags to shorten the full command. As an example, the command above could be written as:

```
docker run -dp 80:80 docker/getting-started
```

The Docker Dashboard

Before going any further, we want to highlight the Docker Dashboard, which gives you a quick view of the containers running on your machine. It provides you access to container logs, lets

Table of contents

The command you just ran

The Docker Dashboard

What is a container?

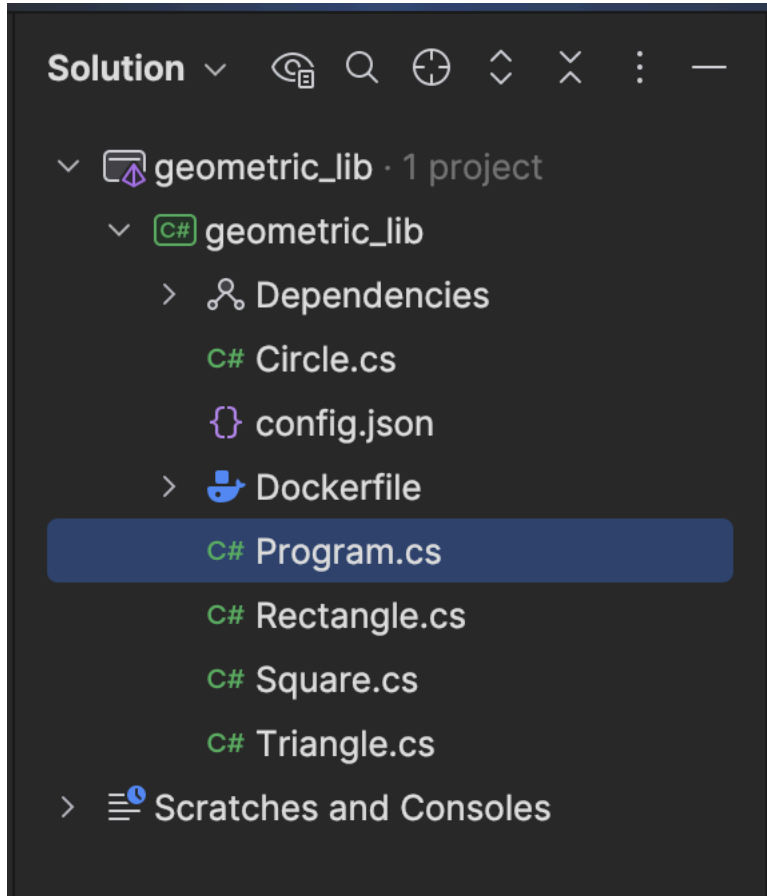
What is a container image?

```
miroslavmeow@MacBook-Air-Miroslav ~ % docker system df
TYPE                TOTAL        ACTIVE        SIZE          RECLAIMABLE
Images              2            2            667.2MB      150.8MB (22%)
Containers          3            2            114.7kB      16.38kB (14%)
Local Volumes       0            0            0B           0B
Build Cache         12           0            749.6kB      749.6kB
miroslavmeow@MacBook-Air-Miroslav ~ %
```

	Name	Tag	Image ID	Created	Size	Actions
	welcome-to-docker	latest	e18e085e051e	10 days ago	443.5 MB	
	docker/getting-started	latest	d79336f4812b	3 years ago	72.86 MB	

3. Создайте docker image, который запускает скрипт с использованием функций из https://github.com/smartigaorg/geometric_lib.

Технология (ASP.NET).



DockerFile

```
# Этап сборки
FROM mcr.microsoft.com/dotnet/sdk:8.0 AS build
WORKDIR /app
COPY *.csproj .
RUN dotnet restore
COPY . .
RUN dotnet publish -c Release -o out --no-self-contained

# Финальный образ
FROM mcr.microsoft.com/dotnet/runtime:8.0
WORKDIR /app
COPY --from=build /app/out .
ENTRYPOINT ["dotnet", "geometric_lib.dll"]
```

2 usages

```
public class Circle
```

```
{
```

1 usage

```
public static double Area(double radius)
```

```
{
```

```
    return Math.PI * radius * radius;
```

```
}
```

1 usage

```
public static double Perimeter(double radius)
```

```
{
```

```
    return 2 * Math.PI * radius;
```

```
}
```

Сборка образа(image)

```
miroslavmeow@MacBook-Air-Miroslav geometric_lib % docker build -t geometric-lib --no-cache .
[+] Building 3.5s (15/15) FINISHED
```

```
miroslavmeow@MacBook-Air-Miroslav geometric_lib % docker run --rm geometric-lib circle 5
=== .NET Geometric Library ===

Data source: command line arguments
Shape: circle
Parameters: 5

=== CIRCLE / ΚΡΥΓ ===
Radius: 5
Area: 78.54
Perimeter (Circumference): 31.42
```

4. Docker Compose (ASP NET MVC + MS Azure)

```
miroslavmeow@MacBook-Air-Miroslav Desktop % git clone https://github.com/zangassis/reminder-list.git
Cloning into 'reminder-list'...
```

```
FROM mcr.microsoft.com/dotnet/sdk:8.0 AS build
WORKDIR /app
COPY ReminderList.csproj .
RUN dotnet restore
COPY . .
RUN dotnet publish ReminderList.csproj -c Release -o out

FROM mcr.microsoft.com/dotnet/aspnet:8.0
WORKDIR /app
COPY --from=build /app/out .
EXPOSE 80
ENTRYPOINT ["/app/ReminderList"]
```

```
appsettings.json
1 {
2   "ConnectionStrings": {
3     "DefaultConnection": "Server=db;Database=ReminderListDb;User Id=sa;Password=YourStrong!Passw0rd;TrustServerCertificate=true;"
4   },
5   "Logging": {
6     "LogLevel": {
7       "Default": "Information",
8       "Microsoft.AspNetCore": "Warning"
9     }
10  },
11  "AllowedHosts": "*"
12 }
```

```

docker-compose.yml
version: '3.8'

services:
  db:
    image: mcr.microsoft.com/azure-sql-edge:latest
    platform: linux/amd64
    container_name: azure_sql_db
    environment:
      - ACCEPT_EULA=Y
      - MSSQL_SA_PASSWORD=YourStrong!Passw0rd
      - MSSQL_PID=Developer
      - MSSQL_USER=SA
    ports:
      - "1433:1433"
    volumes:
      - sql_data:/var/opt/mssql
    networks:
      - app_network

  web:
    build: .
    container_name: aspnet_app
    ports:
      - "9090:80"
    environment:
      - ASPNETCORE_ENVIRONMENT=Development
      - ConnectionStrings__DefaultConnection=Server=db;Database=ReminderListDb;User Id=sa;Password=YourStrong!Passw0rd;TrustServerCertificate=true;
    depends_on:
      - db
    networks:
      - app_network

volumes:
  sql_data:

networks:
  app_network:
    driver: bridge

```

```
[miroslavmeow@MacBook-Air-Miroslav reminder-list % docker-compose up -d --build
```

```

✓ reminder-list-web          Built
✓ Network reminder-list_app_network Created
✓ Volume reminder-list_sql_data Created
✓ Container azure_sql_db      Started
✓ Container aspnet_app        Started

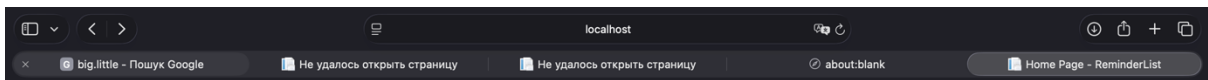
```

```

[miroslavmeow@MacBook-Air-Miroslav reminder-list % curl -v http://localhost:9090
* Host localhost:9090 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
*   Trying [::1]:9090...
* Connected to localhost (::1) port 9090
> GET / HTTP/1.1
> Host: localhost:9090
> User-Agent: curl/8.7.1
> Accept: */*
>
* Request completely sent off
* Recv failure: Connection reset by peer
* Closing connection
curl: (56) Recv failure: Connection reset by peer

```

☐	▼	●	reminder-list	-	-	-	☐	⋮	
☐		●	azure_sql_db	346f443a1eec	azure-sql-e	1433:1433	☐	⋮	
☐		●	aspnet_app	3ff0ff83c79d	reminder-li	9090:80	☐	⋮	



Welcome

Learn about [building Web apps with ASP.NET Core](#).

5. Настроить сети тома для обеспечения связи между контейнерами сохранения данных.

```
[miroslavmeow@MacBook-Air-Miroslav reminder-list % docker volume inspect reminder-list_sql_data
[
  {
    "CreatedAt": "2026-03-12T17:14:37Z",
    "Driver": "local",
    "Labels": {
      "com.docker.compose.config-hash": "ca3be0b41ebbdabaeffb8f47a3eb332ac919009ce431a12e28e7420cd72768fe",
      "com.docker.compose.project": "reminder-list",
      "com.docker.compose.version": "2.40.0",
      "com.docker.compose.volume": "sql_data"
    },
    "Mountpoint": "/var/lib/docker/volumes/reminder-list_sql_data/_data",
    "Name": "reminder-list_sql_data",
    "Options": null,
    "Scope": "local"
  }
]
```

```
[miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network inspect reminder-list_app_network
[
  {
    "Name": "reminder-list_app_network",
    "Id": "1158e0da9ca07694ee3a8a3e5ddef1a1c49688b99b42b19cc2c501c2962a94d5",
    "Created": "2026-03-18T20:17:25.495570418Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "85c85a57117a1dc5a8673850f98246ac08d407a987f63840d627671b5e06524d": {
        "Name": "azure_sql_db",
        "EndpointID": "cab9257cc5116c23839506caff2ccfa8c0bb87d2f61e619f1dccfa2be77f2c49",
        "MacAddress": "26:fe:20:87:bf:79",
        "IPv4Address": "172.18.0.2/16",
        "IPv6Address": ""
      },
      "bfa2f8393129418b20975d1caf025d7e6564a9efd60ea275ffefbc742094475": {
        "Name": "aspnet_app",
        "EndpointID": "16dae166e1977664fd400342b4615ab9e44f9fa2e38c17b735d7d419a14199f1",
        "MacAddress": "8e:06:03:54:ab:5f",
        "IPv4Address": "172.18.0.3/16",
        "IPv6Address": ""
      }
    },
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {
      "com.docker.compose.config-hash": "508cb2a8f4d314a7fee87f039f5bdaa41298020c112026af4f34e21371a05854",
      "com.docker.compose.network": "app_network",
      "com.docker.compose.project": "reminder-list",
      "com.docker.compose.version": "2.40.0"
    }
  }
]
miroslavmeow@MacBook-Air-Miroslav reminder-list %
```


6. Разместить результат в Docker Hub.

```
[miroslavmeow@MacBook-Air-Miroslav reminder-list % docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: SRPL-PRLC
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

Login Succeeded
miroslavmeow@MacBook-Air-Miroslav reminder-list %
```

```
[miroslavmeow@MacBook-Air-Miroslav reminder-list % docker tag reminder-list-web miraslauvitaut/reminder-list-web:latest
[miroslavmeow@MacBook-Air-Miroslav reminder-list % docker push miraslauvitaut/reminder-list-web:latest
The push refers to repository [docker.io/miraslauvitaut/reminder-list-web]
0427f0fac2af: Pushed
199d6be31b25: Pushed
95cc66694fb3: Pushed
628ae826343c: Pushed
b0e683aaa202: Pushed
c8f15a3841dd: Pushed
d997cc310c98: Pushed
733668589fc0: Pushed
0c0e1b37edbd: Pushed
latest: digest: sha256:9a2b75177ada5c0a460faaba734a67abbc0f06325970403f007ce9cc43755f6d size: 856
miroslavmeow@MacBook-Air-Miroslav reminder-list %
```

```
[miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
cd3b84f6df7b        bridge              bridge              local
efb95826db39        host                host                local
3d58a371de56        none                null                local
1158e0da9ca0        reminder-list_app_network bridge              local
miroslavmeow@MacBook-Air-Miroslav reminder-list %
```

7. Изучение особенностей сетевого взаимодействия

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
cd3b84f6df7b        bridge              bridge              local
efb95826db39        host                host                local
3d58a371de56        none                null                local
1158e0da9ca0        reminder-list_app_network bridge              local
```

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network inspect bridge
```

```
[
  {
    "Name": "bridge",
    "Id": "cd3b84f6df7bbcd79949f18b42b2a6f4c2a6077aa02c5a9712027865d33ebd16",
    "Created": "2026-03-18T19:56:51.737422291Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.17.0.0/16",
          "Gateway": "172.17.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {
      "com.docker.network.bridge.default_bridge": "true",
      "com.docker.network.bridge.enable_icc": "true",
      "com.docker.network.bridge.enable_ip_masquerade": "true",
      "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
      "com.docker.network.bridge.name": "docker0",
      "com.docker.network.driver.mtu": "65535"
    },
    "Labels": {}
  }
]
```

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network inspect none
```

```
[
  {
    "Name": "none",
    "Id": "3d58a371de56bc4517e2325ea16cc787ff02f9f6e0f51e11760fcf0041efc37e",
    "Created": "2025-10-21T05:29:59.217409Z",
    "Scope": "local",
    "Driver": "null",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": null
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]
```

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network inspect host
```

```
[
  {
    "Name": "host",
    "Id": "efb95826db39e36e6a25d603590cfecc0e884b0ae32dc7a7cdf3addd67024dbb",
    "Created": "2025-10-21T05:29:59.218799708Z",
    "Scope": "local",
    "Driver": "host",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": null
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]
```

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network inspect none
```

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network create --driver bridge my_bridge_net
```

```
45c7a59c7b115ab35f2459859c521b98a27cd6e10e1e7265508c40c417f05336
```

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
cd3b84f6df7b	bridge	bridge	local
efb95826db39	host	host	local
45c7a59c7b11	my_bridge_net	bridge	local
3d58a371de56	none	null	local
1158e0da9ca0	reminder-list_app_network	bridge	local

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker run -d --name test1 --network my_bridge_net nginx
```

```
c0a161d077d78d3abc2c7c3bb23e5688c5c8ca741acbe4d0e67ad3a6b20e5b5d
```

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network inspect my_bridge_net
[
  {
    "Name": "my_bridge_net",
    "Id": "45c7a59c7b115ab35f2459859c521b98a27cd6e10e1e7265508c40c417f05336",
    "Created": "2026-03-18T20:39:39.841194883Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "c0a161d077d78d3abc2c7c3bb23e5688c5c8ca741acbe4d0e67ad3a6b20e5b5d": {
        "Name": "test1",
        "EndpointID": "fbff5dcfcc8ea7bed265fe765c7bc804277864892b2609b33bc332bcc862bc3f",
        "MacAddress": "e6:03:05:c6:24:e6",
        "IPv4Address": "172.19.0.2/16",
        "IPv6Address": ""
      }
    },
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {}
  }
]
```

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network disconnect my_bridge_net test1
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network inspect my_bridge_net
[
  {
    "Name": "my_bridge_net",
    "Id": "45c7a59c7b115ab35f2459859c521b98a27cd6e10e1e7265508c40c417f05336",
    "Created": "2026-03-18T20:39:39.841194883Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {}
  }
]
```

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker run -d --name c1 --network my_bridge_net2 alpine sleep infinity
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2108c3f659
Status: Downloaded newer image for alpine:latest
b790d9833ba2a664234cfcf386b412d696f48273df59f0aeffa4e02fc980f5b8
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker run -d --name c2 --network my_bridge_net2 alpine sleep infinity
8282e5b7056a0d35a444d006d856dd1afca8e0cf91ddb573b440e28cfe39dfc4
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker run -d --name c3 --network my_bridge_net2 alpine sleep infinity
0220cbec470860aa58c7588e9d067f55019c67b3993297ec9341c3f365a98117
```

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker exec -it c1 sh
/ # ping c2
PING c2 (172.20.0.3): 56 data bytes
64 bytes from 172.20.0.3: seq=0 ttl=64 time=0.841 ms
64 bytes from 172.20.0.3: seq=1 ttl=64 time=0.224 ms
64 bytes from 172.20.0.3: seq=2 ttl=64 time=0.221 ms
64 bytes from 172.20.0.3: seq=3 ttl=64 time=0.287 ms
64 bytes from 172.20.0.3: seq=4 ttl=64 time=0.301 ms
64 bytes from 172.20.0.3: seq=5 ttl=64 time=0.303 ms
64 bytes from 172.20.0.3: seq=6 ttl=64 time=0.282 ms
64 bytes from 172.20.0.3: seq=7 ttl=64 time=0.268 ms
64 bytes from 172.20.0.3: seq=8 ttl=64 time=0.265 ms
64 bytes from 172.20.0.3: seq=9 ttl=64 time=0.333 ms
64 bytes from 172.20.0.3: seq=10 ttl=64 time=0.226 ms
64 bytes from 172.20.0.3: seq=11 ttl=64 time=0.348 ms
64 bytes from 172.20.0.3: seq=12 ttl=64 time=0.425 ms
64 bytes from 172.20.0.3: seq=13 ttl=64 time=0.334 ms
64 bytes from 172.20.0.3: seq=14 ttl=64 time=0.704 ms
64 bytes from 172.20.0.3: seq=15 ttl=64 time=0.260 ms
64 bytes from 172.20.0.3: seq=16 ttl=64 time=0.189 ms
64 bytes from 172.20.0.3: seq=17 ttl=64 time=0.225 ms
ping c3
64 bytes from 172.20.0.3: seq=18 ttl=64 time=0.472 ms
64 bytes from 172.20.0.3: seq=19 ttl=64 time=0.045 ms
64 bytes from 172.20.0.3: seq=20 ttl=64 time=0.252 ms
64 bytes from 172.20.0.3: seq=21 ttl=64 time=0.303 ms
64 bytes from 172.20.0.3: seq=22 ttl=64 time=1.902 ms
64 bytes from 172.20.0.3: seq=23 ttl=64 time=0.308 ms
ex64 bytes from 172.20.0.3: seq=24 ttl=64 time=0.312 ms
it
64 bytes from 172.20.0.3: seq=25 ttl=64 time=0.284 ms
64 bytes from 172.20.0.3: seq=26 ttl=64 time=0.277 ms
64 bytes from 172.20.0.3: seq=27 ttl=64 time=0.270 ms
64 bytes from 172.20.0.3: seq=28 ttl=64 time=0.262 ms
^C
--- c2 ping statistics ---
29 packets transmitted, 29 packets received, 0% packet loss
round-trip min/avg/max = 0.045/0.369/1.902 ms
/ # ping c3
PING c3 (172.20.0.4): 56 data bytes
64 bytes from 172.20.0.4: seq=0 ttl=64 time=0.459 ms
64 bytes from 172.20.0.4: seq=1 ttl=64 time=0.180 ms
64 bytes from 172.20.0.4: seq=2 ttl=64 time=0.179 ms
64 bytes from 172.20.0.4: seq=3 ttl=64 time=0.163 ms
64 bytes from 172.20.0.4: seq=4 ttl=64 time=0.211 ms
64 bytes from 172.20.0.4: seq=5 ttl=64 time=0.262 ms
64 bytes from 172.20.0.4: seq=6 ttl=64 time=0.259 ms
64 bytes from 172.20.0.4: seq=7 ttl=64 time=0.280 ms
64 bytes from 172.20.0.4: seq=8 ttl=64 time=0.283 ms
64 bytes from 172.20.0.4: seq=9 ttl=64 time=0.255 ms
64 bytes from 172.20.0.4: seq=10 ttl=64 time=0.343 ms
64 bytes from 172.20.0.4: seq=11 ttl=64 time=0.253 ms
64 bytes from 172.20.0.4: seq=12 ttl=64 time=0.245 ms
64 bytes from 172.20.0.4: seq=13 ttl=64 time=0.173 ms
64 bytes from 172.20.0.4: seq=14 ttl=64 time=0.428 ms
64 bytes from 172.20.0.4: seq=15 ttl=64 time=0.338 ms
^C
--- c3 ping statistics ---
16 packets transmitted, 16 packets received, 0% packet loss
round-trip min/avg/max = 0.163/0.269/0.459 ms
```



```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network create --driver overlay my_overlay_net
t7m5bs5cwwx0efvcvhl3xi26v
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
cd3b84f6df7b	bridge	bridge	local
1aee6175a4bf	docker_gwbridge	bridge	local
efb95826db39	host	host	local
k2p5mwz1ncat	ingress	overlay	swarm
45c7a59c7b11	my_bridge_net	bridge	local
f8c1aa3f45f5	my_bridge_net2	bridge	local
t7m5bs5cwwx0	my_overlay_net	overlay	swarm
3d58a371de56	none	null	local
1158e0da9ca0	reminder-list_app_network	bridge	local

```
miroslavmeow@MacBook-Air-Miroslav reminder-list % docker network inspect my_overlay_net
[
  {
    "Name": "my_overlay_net",
    "Id": "t7m5bs5cwwx0efvcvhl3xi26v",
    "Created": "2026-03-18T20:46:56.271011502Z",
    "Scope": "swarm",
    "Driver": "overlay",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "10.0.1.0/24",
          "Gateway": "10.0.1.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": null,
    "Options": {
      "com.docker.network.driver.overlay.vxlanid_list": "4097"
    },
    "Labels": null
  }
]
```